

Design Outline for Maximum-Apogee 3D-Printed Rocket (Estes A8-3 Motor)

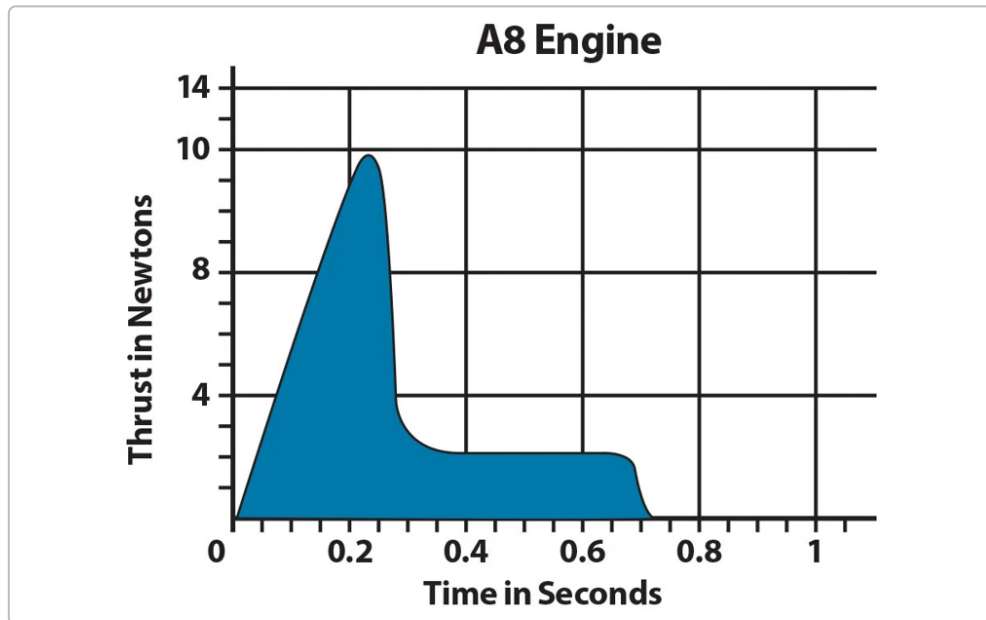


Figure: Thrust curve of the Estes A8 rocket engine, showing its short burn (~0.7 s) and peak thrust of ~10 N ¹. The A8-3 variant includes a 3-second delay for parachute ejection. This brief impulse demands a lightweight, low-drag rocket design to achieve a high apogee.

The following outline details a proposed rocket design to maximize altitude on an Estes A8-3 motor while **meeting all project constraints** (mass < 3 oz, stability 1.10–2.50 calibers, rail exit ≥ 15.24 m/s, safe recovery, 8" print envelope, etc. ² ³). Key design decisions are explained with physical rationale and tied to the given requirements. The rocket will be manufactured from PLA via 3D printing, so considerations for printability, tolerances, and rapid iteration are included. This outline serves as a roadmap for a student team to balance **aerodynamics, stability, weight, and structural integrity** in the design process.

1. Body Tube and Fin Geometry – Minimizing Drag for Altitude

- **Slender Body Tube Diameter & Length:** Use the smallest practical body diameter to reduce frontal area and drag, while still accommodating the 18 mm motor and parachute. A narrower airframe greatly cuts aerodynamic drag, helping to maximize altitude. Keep the rocket as **short as possible while maintaining stability** – a longer rocket shifts the center of pressure (CP) forward (improving stability) but increases surface area and skin friction. Find an optimal length that achieves the needed stability margin without excess length/mass. *Rationale:* Less cross-sectional area means less drag force at a given speed, so a slim rocket will coast higher for the same impulse. However, ensure the internal volume is sufficient for the parachute (see §3) and that the length/diameter ratio still

allows a stable CG–CP relationship. The design will likely resemble a lightweight minimum-diameter rocket built around the A8 motor casing. (**Constraint:** must fit within an 8" × 8" printer, so total length $\leq$ ~8" per part ⁴ ⁵ – the rocket can be in one piece if ~8" long, or in segments if longer.)

- **Streamlined Nose Cone Shape:** Choose a smooth, curved nose profile (such as an **ogive** or **long elliptical nose**) to minimize form drag. For subsonic model rockets, a rounded or ogive nose cone produces the least drag ⁶ . Avoid sharp points (both for safety and because at low speeds, a pointed tip has no drag advantage ⁶) – per the safety requirement, **no overly sharp nose** ⁷ ⁸ . Also avoid a blunt or flat nose, which would dramatically increase pressure drag. A **long elliptical or Von Kármán ogive** nose is ideal for high altitude ⁹ . The nose cone should smoothly blend into the body tube diameter to avoid flow separation. *Rationale:* A curved nose cone shape keeps airflow attached and minimizes pressure drag in the subsonic regime, which our rocket will operate in ⁶ . A well-chosen nose shape reduces energy loss to drag, directly improving apogee. (The difference between a good ogive and a simple cone is modest at low speed, but every bit helps for altitude ⁶ .) The nose tip can be slightly rounded for safety, and the overall length-to-diameter (finess) ratio of the nose might be around 2.5–3:1 for a good drag/stability tradeoff. Ensure the **nose cone has an adequate shoulder** for attachment to the body (shoulder length about equal to body diameter ¹⁰) with proper tolerance ($\approx$ 0.25 mm clearance) so it fits snugly but can pop off at ejection (see §3).
- **Fin Count – Three vs. Four:** Use **three fins** rather than four if stability allows. Three fins are the minimum for stable flight and have lower drag and weight than a four-fin configuration. Each additional fin increases drag and adds mass, which will reduce altitude. While more fins do move the CP aft and increase stability (a four-fin rocket is more inherently stable than a three-fin design), the goal is to stay within the required stability range with the fewest fins practical. With a well-balanced design, three fins at 120° spacing should provide adequate stability in flight. *Rationale:* Every fin produces parasitic drag (skin friction and pressure drag) and adds to the rocket's mass. Three fins reduce these penalties by 25% compared to four fins. Additionally, with fewer fins there is less interference drag (fin-body intersections). Many high-altitude optimized models use three fins for this reason. **Constraint:** Stability must remain $\geq$ 1.10 calibers ² ; if three fins prove insufficient (e.g. CP too far forward), slight increases in fin size or adding nose ballast (last resort) can be used rather than adding a fourth fin. The design will start with three fins to maximize altitude, and simulations will verify the stability margin.
- **Fin Planform and Airfoil:** Design the fins with an **efficient shape and minimal size** (consistent with stability needs). An **elliptical or trapezoidal planform** with the root chord larger than tip chord is ideal, as it reduces drag-inducing tip vortices (elliptical distributions minimize induced drag). Keep the fin span moderate – just enough to ensure stability – and the sweep angle such that the fins present a slender profile to the airflow. Crucially, give the fins a streamlined cross-section: **airfoil the fins** (rounded leading edge, tapered trailing edge) to reduce drag. Even a simple symmetric airfoil or thin wedge shape will significantly lower drag compared to a flat plate. *Rationale:* A properly airfoiled fin can greatly reduce pressure drag and delay flow separation, improving overall aerodynamic efficiency. According to a 3D-printed rocket case study, **airfoiled fins reduce rocket drag and even improve stability** ¹¹ . The fin leading edges should be rounded or at least chamfered, and trailing edges can be thinned to an edge. Also, avoid very thin, sharp tips on the fin planform – better to **clip the fin tips** slightly or use a rounded tip shape to reduce tip drag and make them sturdier (sharp corners tend to break in handling and contribute to vortex drag). Each fin should have a **thickness of ~1.6–3 mm at the root** (the printer's minimum wall thickness is ~1.6 mm ¹² , so this sets a lower

bound for fin thickness for strength). The fins can taper to a thinner section at the outer portions. By keeping fins as small as possible and well-shaped, we minimize drag and mass, directly contributing to higher altitude.

- **Fin Placement and Stability Lever Arm:** Mount the fins low on the body, near the tail, to maximize their moment arm from the center of gravity (this increases their stabilizing effect on the rocket). The **fins should be integrated into the aft body section** (“fin can”) and evenly spaced. Ensure they are perfectly aligned with the body axis (no cant or twist) to avoid inducing roll or drag. The fin root should start a few millimeters above the very end of the body tube – this leaves a small flat rim at the tail for the rocket to sit on (important for printing and for motor installation) and prevents a pointy fin trailing edge from being the lowest point (which could chip). Use **fillets** where the fin meets the body: in the CAD, add a fillet or radius to smoothly blend the fin root into the tube. Fillets reduce interference drag and add significant strength by distributing stress over a larger area. *Rationale:* Aft-mounted fins provide a longer lever arm to the CG, requiring less fin area to achieve the same stability margin. Fillets improve airflow and prevent turbulent separation at the fin-body joint; they also help prevent the fins from snapping off on landing. The placement must also account for the **launch rail clearance** – **do not place a fin directly in line with the launch lug** or rail, to avoid interference ¹³. Typically, if using three fins, one of the gaps between fins is a good spot for the launch lug (placed midway between two fins). Finally, check that with the chosen fin size/placement, the **OpenRocket stability margin is ~1.5 calibers**; adjust fin geometry as needed in simulation before finalizing.

- **Smooth Surface Finish:** Ensure all outer surfaces (body tube, nose, fins) are as smooth as possible to minimize skin friction drag. The 3D printed PLA parts will have layer lines; plan to **lightly sand and/or apply a thin filler/paint** to the exterior after printing to reduce roughness. Aerodynamic drag at the velocities of this rocket (likely ~30–50 m/s peak) is significantly affected by surface finish – smoother surfaces can reduce drag. Also, avoid any protrusions or discontinuities on the airframe that aren’t necessary. For example, recess or countersink the launch lug into a shallow channel or use a small profile lug to reduce its drag profile. *Rationale:* A rough or irregular surface can trip premature turbulence and increase drag. While a fully glass-smooth finish may be beyond the scope of a quick prototype, even a modest improvement in smoothness will help altitude. Additionally, sharp corners should be rounded (e.g., trailing edges of fins slightly sanded) because sharp edges can cause vortex shedding. The **printed layers (0.2 mm)** themselves act like tiny steps – if time permits, the team can sand the leading edges of fins and the nose cone where it matters most. A coat of primer/paint can fill in layer grooves and further smooth the rocket (and make it look “nice,” which is actually encouraged ¹⁴). Prioritize aerodynamics on the forward-facing surfaces (nose, leading fin edges) for smoothing, as these have the biggest impact on drag.

2. Physics-Based Tradeoffs – Mass, Stability, and Drag

- **Motor Impulse and Weight Budget:** The Estes **A8-3 motor provides only ~2.5 N·s of total impulse** ¹⁵, so keeping the rocket’s mass extremely low is critical. The rocket (including the ~16 g motor) must weigh < 3 oz (85 g) per the requirements ¹⁶ ¹⁷, and in practice to reach maximum altitude it should be well under that limit. *Design Decision:* Strive for a **printed rocket mass (without motor) ~50–70 g**, leaving ~15–20 g margin for potential extra hardware (shock cord, parachute, etc.) while staying under 85 g total. A lighter rocket accelerates faster and coasts higher given the fixed impulse. **Tradeoff:** Reducing mass too much can sometimes shift stability (if weight is mainly taken

from the nose, CG moves back) or weaken the structure – we must remove weight smartly (lightweight materials, thin walls) without compromising strength or stability. Use the minimum infill and wall thickness that still yield a safe structure (PLA is fairly strong; 1.6 mm walls and 20% infill are a starting point ¹⁸ ¹²). If the rocket comes out overweight in initial prints, consider trimming length, using fewer perimeters, or cutting non-critical features. *Physics:* For a given impulse, the ideal delta-V scales with $\ln(m_{\text{initial}}/m_{\text{final}})$ (rocket equation). To maximize altitude, we want a small mass ratio – essentially, minimize the rocket’s inert mass since the propellant mass is fixed ~4 g ¹⁹. In summary, **eliminate any unnecessary grams** (shorter screws, minimal glue, no bulky decorations) to allow the limited impulse to propel the rocket as high as possible.

- **Stability Margin (CG–CP Offset):** Target a **static stability margin between 1.1 and ~2.0 calibers** (within the allowed 1.10–2.50 range ²). Aim near the lower end of that range (~1.3–1.5 calibers) for maximum altitude. *Tradeoff:* A rocket just above 1.1 calibers stable will fly straight but not too “over-stable,” which is ideal for altitude in low-wind conditions. If the margin is too low (<1.1), the rocket is unsafe (could become unstable); if too high (>2.5), the rocket is very stable but will **weathercock** aggressively (turn into the wind), resulting in a more horizontal, lower flight path ²⁰. Over-stability also often means extra weight in the nose or oversized fins, which hurt altitude ²¹. Thus, we want **just enough stability for a safe, straight flight**, but not much more. Use OpenRocket to adjust fin area/placement and any ballast such that **CG is ahead of CP by ~1 to 1.5 body diameters** at launch (with motor loaded). Remember that the motor’s weight is in the aft; when it burns, the CG will shift forward, actually increasing stability in flight. So ensure the launch configuration (with full motor) is ≥ 1.1 calibers stable (for launch safety), but it’s acceptable if the margin grows a bit after burnout. *Design choices:* If stability is slightly low, prefer to enlarge fins or lengthen the rocket a tad (which moves CP aft relatively) rather than adding nose weight, since adding weight undermines altitude. Conversely, if simulation shows >2.5 calibers stability, we can **reduce fin size or nose length** to bring it down and save weight/drag. The key is balancing **CP location vs. CG location** – a process of tweaking geometry in simulation to hit the sweet spot. (As a rule, one caliber of stability is a common safety margin in rocketry ²² ²³, which aligns with our lower bound.)

- **Fin Area vs. Drag (Stability Tradeoff):** Increasing fin size (or number) will move the CP rearward and raise the stability margin, but at the cost of **increased drag and mass** ²⁴. This tradeoff must be optimized. *Tradeoff Analysis:* Fins contribute both **pressure drag** (especially if they stall or at angle of attack) and **skin friction**. Larger fins = more drag, which lowers apogee; but too small fins = inadequate stability, risking a dangerous flight. Our approach is to use the **minimum fin area that yields ~1.3–1.5 calibers stability**. For instance, if initial design yields 0.9 calibers, we might slightly enlarge the fins (or forego any reduction in nose weight) to reach stability. If it yields, say, 2.5+, we can shrink fin area to cut drag. We recognize that an optimal design for altitude might be just on the edge of stability, so we’ll design in a small buffer (aim for ~1.3–1.5 to account for any CP calculation uncertainties or slight center-of-mass shifts in reality). *Supporting info:* More fins/area makes a rocket “more stable” in that it corrects its orientation faster in airflow, but in windy conditions that means it will turn into the wind sooner and travel more horizontally ²⁰. A **less stable (but still stable) rocket tends to fly straighter upward**, which is better for altitude especially if there is any wind. Thus, for calm-weather launch aiming at max altitude, we intentionally avoid over-stabilizing. We also consider that an overly stable rocket might require extra nose weight (to move CG forward), which directly subtracts from altitude as dead weight ²¹. In summary, **balance fin size and rocket mass** to achieve the low end of the stability range, thereby minimizing drag and weight penalties.

- Rail Exit Velocity ≥ 15.24 m/s:** Ensure the design can reach at least **50 ft/s off the 1 m launch rail** ³. This is a firm stability requirement – the rocket must be going fast enough when it leaves the guide rail for the fins to effectively stabilize it in free flight. *Implications for design:* Achieving this speed depends on **thrust-to-weight ratio and drag during the first meter of flight**. The A8-3 produces a peak ~ 10 N of thrust and an average of ~ 3 N over 0.7–0.8 s ¹ ²⁵. For an ~ 0.08 kg rocket (80 g), 3 N yields ~ 0.4 g acceleration on average – not enough to reach 15 m/s in 1 m. However, the initial thrust spike ~ 9 – 10 N (about 1.2 g acceleration) will get the rocket moving faster early on. We must keep drag low and mass low so that the rocket **accelerates quickly**. Using OpenRocket or calculations: with a ~ 60 – 70 g rocket plus 16 g motor, we expect to clear 15 m/s in roughly 0.3–0.4 s of engine burn (covering ~ 1 m distance). If the rocket is heavier or draggier, it might leave the rail too slowly (< 15 m/s) and risk weathercocking or instability. *Design strategies:* **Minimize liftoff weight** (as above) and ensure a **clean launch configuration** – e.g., the launch lug should be not too long and aligned to avoid friction on the rod. A small angle of attack on launch due to wind can slow the rocket, so if anything, lean toward slightly **more stability** (≈ 1.5 – 2.0 calibers) if launches might be windy, to correct quickly. But primarily, keep mass low. The **motor itself is ~ 15.6 g** ²⁶ and the propellant burns fast, meaning the rocket's average acceleration is highest at ignition and then dropping. We want the rocket to be as light as possible at lift-off to take advantage of the thrust spike. Also, any reduction in drag (smooth surfaces, smaller diameter) means less resistance during that crucial rail acceleration phase. This is a direct reason to maintain a small diameter and smooth fins. We will verify in simulation that the **rail exit velocity ≥ 15.3 m/s** (satisfying the requirement) ³, and iterate the design if not. Typically, if the criterion isn't met, the fixes are: reduce weight, reduce drag (e.g., smaller fins or better nose shape), or in a real scenario use a longer rail – but our rail is fixed ~ 1 m per spec ²⁷, so we optimize the rocket instead.
- Motor Delay and Apogee Timing:** The A8-3 has a **3 second delay after burnout** before the ejection charge fires ². We must design for the rocket to **reach apogee around that 3–4 second mark** so that the parachute deploys just after apogee (when the rocket has started to tip over). *Tradeoff:* If the rocket coasts too high for too long, apogee might occur well after 3 seconds – meaning the ejection charge would fire while the rocket is still climbing, possibly causing a high-speed deployment (risking parachute damage). If the rocket is too heavy/low-performance, it might peak earlier (e.g. 1–2 seconds after burnout) and then be falling for 1+ second before ejection – which is dangerous because the parachute would deploy during a fast descent. We want the **ejection to occur very near apogee** (preferably just after). *Design influence:* This is a subtle consideration, but it affects how we might choose our target altitude and coast duration. A lighter, slick rocket could potentially overperform such that 3 seconds is too short (though with an A8's limited impulse, this is less likely). More commonly, if our rocket is a bit heavy (~ 80 g) the apogee might occur about 2–2.5 seconds after burnout. This would mean the ejection charge at 3 seconds happens slightly after apogee (which is okay as long as it's not too late). According to the requirement, we must **ensure ejection happens after the anticipated apogee** ², implying apogee should occur before ~ 3 seconds post-burnout in simulation. If our design shows apogee at, say, 2.0 seconds after burnout (2.7 s after launch) and ejection at 3.7 s, that's acceptable – the rocket will be descending only a short time (ensuring chute opens when there's some airspeed to inflate but not so much to shred it). On the other hand, if a design iteration gave apogee at 4+ seconds after burnout, that's a problem (ejection would be too early, while still climbing). In that case, ironically, one might need to *add* a bit of drag or weight to shorten the coast (though with A8 that scenario is unlikely unless the rocket is extremely light < 40 g). Conversely, if apogee is too early (< 2 s), we might try to lighten or streamline the rocket to extend the coast slightly. This interplay will be checked in OpenRocket: we will adjust the design to

target an apogee just before 3 seconds after motor burnout, fulfilling the “ejection after apogee” requirement ². In summary, **coordinate the flight profile with the fixed 3 s delay** so that parachute deployment is well-timed.

- **Parachute Size vs. Descent Rate:** For recovery, the rocket must **land safely with low speed** – the goal is ≤ 5 m/s landing velocity (to avoid damage) ²⁸. A larger parachute gives a slower descent but creates more drag during ascent (slightly reducing altitude). We must choose a parachute diameter that balances these factors. The assignment suggests a ~30 cm (12”) parachute, which when properly deployed should bring an <85 g rocket down around or under 5 m/s, especially with some air resistance from the rocket body. *Tradeoff:* A 30 cm chute is likely appropriate; going significantly smaller would risk a hard landing >5 m/s ²⁹, while larger could be overkill and harder to stow. However, it’s critical that whatever chute we choose **fits in the rocket (volume ≥ 15 cm³) and deploys reliably** ³⁰. We might consider a slightly smaller chute if volume is tight, combined with adding a spill hole to reduce oscillation and ensure full inflation. The **descent speed** scales roughly with the inverse of chute diameter (for the same weight), so even a 25 cm chute might yield ~6–7 m/s which could be borderline for landing on hard ground. Since safe recovery is a must, we will err on the side of a generous chute and just account for its packing in the design. The small extra drag of a packed parachute during thrust is minimal (it’s inside the rocket), and once ejected at apogee, any additional drag only helps slow descent. So, **for flight performance, parachute size has little negative effect until deployment**. The key is purely to ensure slow landing. Therefore, we plan for ~30 cm nylon chute, expecting a descent rate ~4–5 m/s. If testing shows a faster descent, we’ll increase chute size or add a second chute or streamer. This decision interacts with design in that the **body tube diameter/length must accommodate the parachute** (see next section), and the shock cord must be long enough to allow the chute to catch air fully.

3. Ejection and Recovery System Integration

- **Parachute Bay Volume:** Allocate sufficient space in the rocket for the parachute and recovery wadding. The design must include a **parachute compartment of at least 15 cm³** (cubic centimeters) as recommended ³⁰. In practical terms, for a ~30 cm diameter parachute, aim for roughly a segment of the body tube about 5–6 cm in length (if internal diameter ~25 mm) to store the folded chute and wadding. The **internal diameter of the body tube** will be only a bit larger than the motor (which is 18 mm); if we stick to a minimal diameter (~20 mm ID), 15 cm³ would require ~an 8–10 cm length of space. This is a significant portion of the rocket’s length, so design the rocket’s internal layout with a clear **parachute bay** above the motor mount. If needed, slightly widening the body tube in the parachute section could help (e.g., a short transition to a slightly larger diameter compartment), but that adds drag – likely we keep a uniform diameter and ensure length is enough. *Design features:* The parachute bay should be free of obstructions and **smooth-walled**. Any internal ledge (like at a coupler joint) should either be sanded flush or be located below the parachute so it cannot snag the canopy. We also ensure **venting**: while the motor’s ejection charge will pressurize the tube to push out the nose and chute, having a mostly hollow volume with some wadding is fine. The earlier checklist item ensures “ ≥ 15 cm³ of space left for parachute” ³¹ ³⁰ – we will meet or exceed this. If our rocket ends up very narrow and it’s challenging to fit the chute, we might either reduce the chute size slightly or use a thinner material chute. However, given safety, we prefer to keep the chute size and adjust the rocket diameter if absolutely necessary (for instance, using a **BT-50 (~24 mm ID)** body instead of BT-20 (~18 mm ID) for more volume, at the cost of a bit more

drag). The parachute bay design will also consider **ease of packing** – a slightly longer bay makes packing easier and reduces the chance of a stuck parachute, so we won't shortchange this length.

- **Nose Cone Attachment and Ejection Fit:** The nose cone will double as the **ejection plug** – it must fit snugly enough not to fall off during launch under drag, but loose enough to blow out when the ejection charge fires. Achieve this by designing a **shouldered nose cone** that inserts into the body tube with a **~0.25 mm clearance** on each side (a total of ~0.5 mm smaller in diameter than the body's inner diameter) ³². This tolerance has been suggested for body tube-shoulder fit and usually provides a nice **"friction fit"**: the nose stays in under its own weight and minor jostling, but a moderate push (or internal pressure from ejection) will pop it off. *Design specifics:* The nose cone's shoulder length should be about equal to the body diameter (e.g., ~20–25 mm long) ¹⁰ to ensure alignment. We will likely **not use any locking or twist mechanism** – friction/tape is sufficient and safer for deployment. We must also heed the note: "don't make the seal too tight" ³³ – if the nose is too tight, the ejection charge might not be able to push it out, leading to a failed deployment (and a ballistic rocket coming down). Therefore, we will test the nose fit: it should **slide out with a firm push** and perhaps **hang on only lightly if the rocket is inverted and shaken** (with parachute in, usually the friction of the folded chute helps hold it too). If it's too loose, we can add a bit of tape on the shoulder to tighten it on launch day; if too tight, we'll sand it down. The ejection charge of the A8-3 is a **small black powder charge** that creates a burst of pressurized gas ³⁴. To make sure this gas can do its job, the **motor mount must not fully seal the tube** (next bullet covers motor block hole), and the nose must be able to vent that pressure by coming off. *Recovery sequence:* at ~3.7 s after launch, the delay burns through, the charge pops – the nose should **separate cleanly**, tethered by the shock cord, and the parachute then is pulled out by the force of the nose cone and expanding gases. We design the nose so that when it's seated, there is **no gap** (to avoid drag) but also no physical lock – just friction. We also avoid painting the shoulder or anything that could glue the nose in place. Finally, recall the requirement to avoid sharp nose tips ⁸ – we'll have a slightly rounded nose tip, which also slightly reduces the chance of the nose getting jammed in the tube due to imperfections (sharp tips can cause misalignment if bumped).

- **Motor Mount & Motor Block:** Implement a robust **motor mounting system** that holds the Estes A8-3 in place during thrust, and positions it correctly relative to the body. The motor is 70 mm long ³⁵; we will design the motor mount tube as part of the fin can or main body with an **internal diameter of ~18.3–18.5 mm** to fit the motor plus a bit of clearance (0.25 mm radius clearance as with other parts) ³². **Motor block:** We will include a **bulkhead (thrust block)** near the forward end of the motor to prevent it from shooting forward into the rocket. According to guidelines, "upon ignition, there needs to be a retention system to prevent the motor from pressing into the body tube" ³⁶ – this bulkhead serves that purpose by physically blocking the motor. We'll place the motor block slightly less than 70 mm from the bottom of the rocket, so that when the motor is inserted from the aft end, **about 5–10 mm of the motor sticks out the back** ³⁷. This exposed bit allows us to grip and remove the motor after flight (motors often expand slightly or get sooty, so easy removal is important). The **motor block will have a central hole** for ejection charge gases ³⁸ – we'll design, say, a 6–8 mm diameter hole (80–90% of the ID as recommended) so that the black powder charge can freely send gas into the parachute bay. The block itself can be a ~2 mm thick ring or ledge inside the tube, printed as part of the fin can. If the rocket is printed in two parts (fin can + upper body), the motor block might double as the interface floor between them. We'll ensure it's strong (maybe a solid layer or two in the print). *Motor retention:* * *The motor block stops the motor from moving upward; we also must ensure the motor doesn't fall out the back or get ejected downward by recoil. Typically, hobby*

rockets use a hook or tape. We have two retention considerations: during thrust, friction is usually enough to keep the motor from sliding back, but tape can be added around the motor for snug fit. After burnout, the ejection charge actually pushes forward on the motor (Newton's third law to push the nose out) and can kick the motor backward out of the rocket if not restrained. To prevent a "spit motor," we will likely rely on friction/tape or design a small lip or cap at the aft end. One idea: a threaded cap that screws onto the motor mount end (though that's complex for a student project and adds weight). Simpler: drill a small hole and use a pin or a wrap of tape across the motor and onto the body to hold it. Given available materials (tape, hot glue), a wrap of masking tape around the joint between motor and body could suffice. Since we want easy motor change, we won't permanently glue it. Another printed option: a slight inward ridge at the aft end of the motor mount that the motor's lip catches on – but Estes A8 doesn't have a big lip like larger motors, just a small crimp. We might rely on tape friction primarily. We will verify in testing that the motor stays in place when the ejection fires (perhaps by static testing the ejection charge or inspecting post-flight). Summary: The motor mount design centers the motor, provides a forward stop (block with a hole), and a means to secure the motor from falling out, all while allowing easy removal. It also thermally isolates the motor (somewhat) from the plastic body – the guidelines mention keeping the PLA from melting ³⁹, so our design might include a cardboard liner* in the motor tube or at least not have very thin plastic right against the nozzle. We can use a bit of cardboard (an allowed material ⁴⁰) inside the motor tube as a sacrificial layer if needed, or simply rely on the short burn time and moving air to not melt anything (A8's aren't too hot, but caution is noted).

- **Launch Lug Integration:** Incorporate a **launch lug (guide tube)** on the rocket's exterior, positioned near the rocket's balance point. The launch lug should have an inner diameter of **~3.81 mm (0.15")** to fit a standard launch rod (likely 1/8") ⁴¹. We will either print the lug as a small tube piece and attach it, or integrate it into the body model. The lug should be about **1/2" (12-13 mm) long** ⁴¹ – long enough to guide but short enough to minimize drag. **Placement:** Mount the lug **close to the center of gravity (CG)** of the loaded rocket ²⁸. This ensures the rocket doesn't bind at the end of the rail and that the rail imparts a straight guidance. Likely, CG will be somewhere around mid-body; we can place one lug roughly at mid-body (or use two smaller lugs spaced apart for more stability on rail, though on such a small rocket one is usually sufficient). We must be careful not to put the lug where it interferes with a fin. The guideline explicitly says "*launch lug is not directly on top of fins*" ¹³ – so we'll position it in between fin positions, perhaps offset 45–60° from a fin if using three fins. Ensure the lug is perfectly parallel to the rocket's axis; any skew can cause friction on the rod. **Aerodynamics:** The lug does add a bit of drag (and even stability effect, though minor). To mitigate drag, keep the lug diameter just large enough and the length minimal. Optionally, we can shape the ends of the lug to a tapered profile or even consider cutting it in half lengthwise (some rocketeers do this to reduce drag by making a semi-tubular guide) – but given our focus on reliability and printing, a full small tube is fine. We could also decide to **use a launch rail button** (small hardpoints) instead of a lug, but since the assignment provides lug specs, we'll stick to a tube. **Manufacturing:** If printing the lug integrated, ensure the orientation can print it without extreme overhangs (maybe print the rocket vertically so the lug prints as a vertical tube – might need support unless aligned with Z axis). Alternatively, print the lug separately (horizontally for strength) and glue it on. Given that attaching after allows alignment using an actual rod inserted (ensuring it's straight), that might be prudent. We will design a flat on the body or a small standoff where the lug goes, to ease alignment. Summing up, the launch lug will be **small, light, properly located, and securely attached**, enabling the rocket to launch smoothly and meet the rail exit speed requirement.

- **Shock Cord and Attachment Points:** Use a **Kevlar shock cord** to connect the nose cone, body, and possibly the motor mount (so that no parts fall independently). The shock cord will likely be anchored at two points: one on the nose cone's inside base and one on the fin can or upper body section. The guidelines require "two bars or hooks (≥ 0.3 cm thick) to attach shock cords: one in the body tube and one in the nose cone" ⁴² ⁴³ . We will integrate these attachment features into our design. For example, inside the nose cone, we can print a small **cross-bar or loop** across the interior near the base of the shoulder. It should be at least 3 mm thick to be sturdy ⁴³ . A hole can be provided if using a loop, or we simply tie the Kevlar cord around the bar. In the body, a similar bar can be placed either at the forward end of the parachute bay (just below where the nose shoulder sits) or as part of the fin can (somewhere accessible from the top opening before the rocket is fully assembled). One design idea is to make the rocket in two pieces (fin can + upper body) so that before joining them, you can tie the shock cord onto an internal bar in the fin can easily ⁴⁴ . Then the upper body slides on (coupler) and perhaps is glued, trapping the cord. If the rocket is one-piece, we'd have to reach in from the top (which for a narrow tube ~20 mm can be tricky) to tie onto a bar near the bottom – hence the two-piece approach is favored for assembly ease ⁴⁴ . The **shock cord length** should be around 2–3 ft (~60–90 cm) for a small rocket – long enough to absorb the shock of deployment and keep nose and body separate without snapping back. We'll likely attach the parachute about 1/3 of the way down the cord from the nose (common practice to reduce entanglement: nose – 1/3 cord – chute – 2/3 cord – body). *Ensure no loose parts:* The cord must connect **all major parts** so that nothing falls freely if separated ⁴⁵ . That means the nose and the rest of rocket at minimum. The motor itself ideally stays in the rocket or is caught (though with a small motor, if it falls out it's light; still, standard practice is the motor stays in the rocket or in the mount). We will assume the motor stays in by retention. If not, one could theoretically tether it too, but that's uncommon and not required if it's retained. *Robustness:* Kevlar is fire-resistant, which is why it's specified (the ejection charge won't burn it). Make sure any **plastic attachment points are beefy** because the shock load at ejection can be high (the guideline of ≥ 3 mm thickness for anchors ⁴² is to ensure they don't snap). We might even do a test by pulling hard on the cord after attaching to see if the anchors hold. If a printed hook seems weak, we'll redesign as a full cross-bar. The attachment points should also be placed such that they **do not interfere with parachute packing**. For example, the body's anchor bar could be low in the parachute compartment, but then the parachute has to slide past it – better to put it up high, just inside the open end of the tube or on the coupler, so that all cord and knots pull out easily. Similarly, the nose's anchor should not protrude in a way that snags fabric. Recessing them slightly or smoothing their edges will help. In summary, we will incorporate **reliable, simple shock cord anchors** in nose and body, and use a length of Kevlar string (e.g., 150 lb test is common) tied securely (likely with a figure-eight knot or double knot and maybe a dab of glue for security). This ensures that when the ejection occurs, the nose comes off but stays attached, pulling the parachute out, and the rocket components remain tethered together during descent.

- **Parachute Installation and Wadding:** The **parachute**, likely made from lightweight nylon cloth (provided material ⁴⁶), will be approximately 12" in diameter with spill holes as needed. It should be attached to the shock cord, typically by a loop or swivel. When packing the rocket, we'll insert a couple of **squares of flame-resistant wadding** (provided ejection wadding ⁴⁷) into the body above the motor (but below the parachute) to protect the chute from the hot ejection gases. The design must leave room for this wadding in addition to the parachute – but wadding can conform to spaces. (Our ≥ 15 cm³ allowance covers parachute; we assume wadding will just fill nooks around it.) *Integration:* Make sure **nothing blocks the parachute's path** out of the rocket ⁴⁸ . This includes

ensuring the nose cone shoulder isn't so long that it restricts space or that any step in diameter inside won't trap the folded chute. If the rocket uses a coupler between fin can and body, perhaps design that coupler to also act as a pseudo "piston" or just ensure its forward end is smooth so the chute slides out. Possibly bevel the inside edge of the body tube at the top to ease the chute's exit. Also consider **parachute folding and placement**: It often helps to have the shock cord loosely folded on top of the chute or vice-versa so that as the nose comes off, it pulls the cord which in turn pulls the chute out. We will recommend using **baby powder** on the chute as a packing aid to prevent it sticking (static). *Parachute reliability*: Given this is a single-parachute design, we must be confident it will deploy. The design already addresses the nose ejection force (hole in motor block, not too tight nose) and anchor points. We should also make sure the **parachute is not too tightly packed** – the tube diameter and length must allow a gentle packing. If we find in testing that the parachute is packing very tight, we might reduce chute size by a small amount (or use a thinner material/chute folding technique). However, since safe recovery is critical, we would rather **increase body tube diameter** slightly to fit the chute than shrink the chute too much. Finally, ensure **no entanglement**: keep the interior of the rocket free of any protrusions or rough areas that cords can catch. The shock cord should be attached in a way that the knot tails won't snag (we can trim excess from knots and maybe tuck them). All these design choices aim to fulfill the requirement that *"parachute ejects successfully – there can't be anything blocking the ejection"* ⁴⁸ and that assembly of the recovery system is not overly fiddly (don't require tying a knot in a deep, narrow tube at final assembly – plan the attachment conveniently) ⁴⁹.

- **Recovery Safety Considerations**: The rocket's recovery system as designed will ensure **no loose debris** (the motor is retained, and nose cone is attached via shock cord) so nothing can fall separately that could injure or be lost ⁴⁵. The parachute will bring the rocket down to an impact under ~5 m/s, minimizing damage to the rocket and ground. Also, a note on **nose cone shape** as it relates to recovery: since occasionally chutes fail, the guideline of "no sharp nose" ⁸ ensures that even in worst case, a lawn-dart impact is less hazardous. Our rounded nose supports that safety aspect. We also painted fillets and rounded fin edges partly for aerodynamics, but it too improves safety (no sharp fin tips to stab anything on landing or handling). In preparation for launch, we will instruct the team on **proper folding of the chute**, insertion of wadding (at least 2-3 sheets or as needed to fully protect the chute), and checking the shock cord for any signs of wear. The integration we designed should make these steps straightforward (e.g., a two-part body makes it easier to lay the chute in without compressing it too much, if we go that route). In summary, the recovery system is robust and fully integrated: it meets the class constraints and will reliably recover the rocket so it can be launched multiple times if needed.

4. Structural and Printability Considerations (PLA 3D Printing)

- **Modular Design & Print Volume**: Plan the rocket as **multiple printable sections** to fit the printer and to ease printing. No single part can exceed 8"x8"x8" ⁴, so if the rocket is, say, 10" tall in total, we might split it into two segments (e.g., a fin can + lower body, and an upper body with nose). Even if the rocket is small enough to print in one go (~6-7" long), splitting into sections can still be beneficial: it allows printing complex geometries in optimal orientations and makes internal features (like shock cord bars or motor block) easier to create and access. A likely modular breakdown is:
 - **Nose Cone with integrated shoulder** (print as one part).
 - **Main body tube** (could be combined with fin can or separate).
 - **Fin Can (aft body with fins and motor mount)** – potentially combined with the lower body tube.

- **Couplers/Shoulders** for joints (if body is in multiple pieces).

Each joint between parts will have a **coupler piece or built-in shoulder** of length roughly equal to the body diameter (for strength and alignment) ¹⁰. For instance, if the body tube and fin can are separate, one of them will have a coupling extension that slides into the other. We will use the recommended **0.010" (~0.25 mm) clearance** on radius at these interfaces for a snug fit that can be glued or taped ¹². It's critical to test these fits with small prints. The couplers might be printed as part of one section or as separate rings to be glued in; integrating them into the model ensures perfect alignment but printing separate can sometimes allow finer tuning. We will ensure **flat mating surfaces** at joints for even gluing. *Example:* The nose cone's shoulder is effectively a coupler into the body tube. The fin can might have an upper coupler that goes into the main body tube section above it. If the rocket is small ($\leq 8"$), we *could* print it as one piece, but internal shock cord hooks and motor block might be unreachable to finish assembly. The hint given suggests splitting fin can and body to tie shock cord easily ⁴⁹, so we'll follow that approach. This modular approach also aligns with rapid prototyping – if one part fails or needs redesign (say fins snapping), we can reprint just that section rather than the whole rocket. - **Printer constraints:** Keep components within ~200 mm (8") in any dimension. Also, ensure no part has features that exceed the print bed size when oriented appropriately. For example, fins sticking out should be accounted for in XY plane. We will double-check that the fin span plus body diameter fits in 8" bed – likely fine, as even a 50 mm fin span + 25 mm body = 75 mm ~ 3" total radius, way under 8".

- **Optimal Print Orientations:** Orient each part to **minimize supports** and ensure strength in critical directions. General guidelines: **print parts vertically** when they are primarily rotationally symmetric (like body tubes, nose cones) so that layer lines run horizontally around the circumference, giving good hoop strength and a smooth axis. Print parts **flat on a stable base** whenever possible ⁵⁰. For our rocket:
- **Nose Cone:** Print **point-up, base-down** on the build plate. The base (shoulder end) is flat and provides a good foundation. As it prints upward, the curved nose will come to a point. Without support, a sharp nose point might sag, but since we will blunt it slightly, it should print fine. The overhang of the nose's slope is gradual (ogive shapes are $\leq 45^\circ$ for a long portion, which might print without support; if it's more conical, it's still usually okay). We avoid a perfectly pointed tip since that's not allowed (and it's unprintable anyway because it'd be one thin spike). We can always sand the tip round if needed. If the nose includes an internal shock cord bar, we might orient that bar vertically or at an angle so it prints as part of the model without overhang – e.g., a bar spanning the interior horizontally would require support inside, which is undesirable. We could instead incorporate a **loop hole** that goes through the shoulder (like a tunnel) or a diagonal beam, which the printer can handle with minimal support. This may need some creativity in CAD (or we print the nose in two halves and glue, but that's less strong and more work – not ideal unless absolutely needed for internal features).
- **Body Tube:** If printing as a separate cylindrical section, the best orientation is **vertical** (standing up) so that the circle cross-section is perfectly round (layer lines make nice circles) and the tube is strong circumferentially. Printing a tall thin tube can cause wobble, but our tube is short (maybe a few inches) and we can use a brim for stability. Also, vertical printing means the inside is automatically hollow, no supports needed for the hollow center, and any small holes (like for venting or payload maybe) would print fine on vertical walls. The layer lines will run hoop-wise, meaning the tube is weaker in the vertical direction (split between layers), but primary stresses (pressure from ejection, bending along length) should be manageable. We will ensure a few perimeter shells to give it strength.

- **Fin Can + Motor Mount:** This is the trickiest. One approach: **print the fin can vertically**, base-down. The flat base would be the bottom of the motor mount (perhaps with a small flat ring around where the motor protrudes). The fins would then print sticking out horizontally initially – which is problematic because a horizontal fin would require support under it until it reaches a 45° angle. To address that, we can design the fins with a slight **cant or airfoil** such that the first part of the fin coming off the body is at a 45° or steeper angle. For example, fillet the fin's root leading edge into the body so it gradually emerges. This way, the overhang might be kept <45°. If that's not sufficient, we may accept some support material under the fins for the first few mm of print. Another strategy is to split the fin can into two halves lengthwise (along the rocket's axis), lay each half flat (fins in plane with bed), print them, and then join them. But that introduces a seam and alignment issues, so we prefer one-piece if possible. The requirement "all components should have a flat bottom to sit on the print bed" ⁵⁰ hints that maybe we design the fin can so that when placed on the bed, the fins' bottom edges and the aft ring all lie in the same plane. I.e., the bottoms of the fins are co-planar with the base of the rocket. If we make the fin trailing edges not go below the rocket's base plane, then the rocket can literally stand on its fins on a table. However, truly flat fins at the bottom would be horizontal – again an overhang to print. But if the fins are clipped (bottom part is flat and parallel to base) and only the upper portions curve out, the very first layers of the fins would just be outlines on the bed extending from the body – these could sag if not supported. Perhaps small sacrificial support ribs or a skirt could be used. We will likely have to experiment with orientation. It might end up that **printing the fin can at a slight angle** (e.g. 30° off vertical) with support is best to get a clean fin surface. But since we have time for iteration, we might try the straightforward vertical print with supports under fins initially. PLA supports break away, and we can sand the underside of fins a bit. The rest of the fin can – motor tube, internal structures – will print fine in this orientation (vertical).
- **Launch Lug:** If integrated into the body print, ensure it doesn't require support internally. We could design the lug as a half-tube on the surface (flat on one side) so that it prints as part of the wall without needing a cylindrical support. Or print the lug separately flat (best strength along its length) and glue it on.

In all cases, maintain the **45° overhang rule** ⁵¹ : any design feature that would create an overhang steeper than 45° from horizontal will need support. Examples to avoid or mitigate: horizontal interior ledges (e.g., shock cord bars – can angle them), holes on top surfaces (might sag – but a small hole is fine), the underside of a nose cone if printed nose-up (the shoulder cavity might need support unless we incorporate a chamfer). We will add **chamfers or fillets** to edges that would otherwise create unsupported overhangs. For instance, the aft end of the motor tube can have a chamfer so it's not a straight 90° overhang. Also, **orient open ends upward** so that the printer can bridge them gradually. The design will emphasize geometry that prints cleanly to reduce post-processing and print failures.

- **Minimum Wall Thickness and Infill:** Adhere to a **minimum wall thickness of ~1/16" (1.6 mm)** for strength and printer capability ¹² . Our CAD will enforce any thin sections (like body tube walls, fin edges) to be at least this thick (which typically equals ~2 perimeter lines at 0.4 mm nozzle). This ensures the rocket is not too flimsy and that the printer can actually produce the walls. Use **2-3 perimeter shells** in slicing to get a nice strong tube and fin edges. Infill can be set to ~20% (the guideline) ¹⁸ or possibly less if weight is a concern and the structure doesn't need it. Many parts of a model rocket can be printed effectively hollow with just skin (like body tubes) because the loads are low. But for areas like where fins attach or the couplers, some infill ensures solidity. We might use different infill for different parts: e.g., nose cone could be mostly hollow (we might even print it in "spiral vase" mode with single wall if weight is crucial, then add a bit of clay for ballast if needed). The

fin can might have higher infill (for strength at fin roots and around motor mount). We will also consider **infill patterns**: a lightweight pattern like triangular or hexagonal provides good rigidity for the weight. *PLA properties*: PLA is strong in compression and fairly stiff, but brittle in flexure. Thus, it's good to have a little give – a slightly lower infill (making parts semi-hollow) means they can deform and absorb impact rather than crack. For example, fins printed solid might crack on landing, whereas fins with sparse infill might flex a bit. However, if fins are too hollow, they could crush or split if you accidentally step on them. We'll aim for a balance: likely 2 shells and 15–20% infill is fine for fins and body. If the rocket is coming out overweight, we'll revisit these settings. The **nozzle diameter (0.4 mm)** and layer height (0.2 mm) mean we shouldn't design any detail smaller than ~0.4 mm. The guideline says any geometry smaller than nozzle (~0.2 mm) won't print ⁵² – so avoid super fine features. Our design should reflect that (no tiny vent holes less than ~1 mm, no thin fin trailing edge below ~0.4–0.5 mm thickness, etc.). In summary, use **reasonable thickness throughout**: body tube walls ~1.6 mm, fin bases maybe 2–3 mm and tapering, nose cone tip maybe 1–2 mm thick if hollow. All this keeps the rocket light but strong enough.

- **Tolerances and Fitment**: All interacting parts need built-in clearance to account for printer tolerances. We will use the recommended **0.25 mm clearance on each side (0.5 mm total)** for couplers, shoulders, and motor fit ¹². For example, the motor diameter is 18.0 mm nominal; we'll make the motor tube ~18.5 mm internal diameter. After printing, we'll test with an actual motor – it should slide in snugly. If it's too tight (maybe material shrinkage or printer minor error), we have that 0.5 mm buffer to sand or we can wrap motor in tape if too loose. Similarly, nose shoulder vs body ID: aim for ~0.3 mm total clearance so it's not wobbly but not sticky. The couplers between body sections: maybe a tad more clearance (~0.5 mm total) since they might be longer engagement and we may glue them (glue needs some gap). The launch lug inner diameter of 3.81 mm is given presumably as final requirement ⁴¹; we should make it slightly larger (maybe 4.0 mm) because any slight sag or blob inside could shrink it. Better to sand or just ensure rod slides freely without binding. Also, we need to consider **tolerance for the shock cord through any holes** (if we design a hole for the cord, make it a few mm wide to easily thread the cord and knot). In assembling, if parts are too tight, we have sandpaper (allowed) to adjust ⁵³. If too loose, we can use a wrap of masking tape or a dab of glue to tighten – but ideally our prints will be close. **Motor fit caution**: The requirement explicitly says “*MAKE SURE YOUR MOTOR FITS*” ⁵⁴ – so we will double-check that dimension by printing a small tube section early on. PLA tends to shrink ~0.2–0.4%, so designing in that 0.5 mm clearance is usually enough. We should also ensure the **motor block alignment** is correct (if it's slightly off center or tilted, the motor might bind – so print it as part of the fin can to ensure perfect concentric alignment).

- **Structural Reinforcements**: Use fillets, bridges, and ribs in the CAD to strengthen key areas without excessive weight. For example, at the fin roots, we will model a **fillet or triangular gusset** to distribute stress from the fin into the body. This not only helps aerodynamically but also means there's more material anchoring the fin. Inside the fin can, around the motor mount, consider **small ribs** connecting the motor tube to the outer body for rigidity (though in a small 18 mm rocket, the entire body might just be the motor tube). If using a slightly larger body tube (like 24 mm) around an 18 mm motor, definitely include centering rings or ribs to hold the motor. But if our rocket is basically 18–20 mm diameter, the motor pretty much touches the walls, so less worry there. **Layer adhesion**: design to avoid stress solely along layers. For instance, fins printed upright will have their layers running spanwise, which is a weak orientation for bending (they could snap along a layer line upon landing impact). To mitigate, we rely on fillets (which give more cross-sectional area) and possibly

consider printing fins at a slight angle or with a thinner profile (less bending force). Some builders might print fins separately flat for strength (strong in X-Y plane), then attach – but since we integrate, we make do with fillets and sufficient thickness. We might also use a **“pause print” technique** to insert something (like a fiberglass veil or carbon rod) into a fin mid-print for strength – but that’s likely overkill for this project.

Additionally, account for **landing stress**: The rocket may land on its fins or tail. The design should ensure fins can handle that. The PLA fillets help, and perhaps slightly thicker fin trailing edge or a tiny nub at fin tip (so it contacts ground slightly before a sharp edge would) could help. Also, our parachute is meant to slow it to ~5 m/s, so impact is mild. We should still ensure the **fin attachment is robust** – maybe incorporate a small internal lip where the fin meets body to increase bonding. Printing in one piece already means it’s monolithic (no glue joint to fail), which is good.

- **Thermal and Safety Considerations:** PLA’s low softening temperature (~60°C) means extended exposure to motor exhaust or ejection charge can deform it ³⁹. In our design, the **motor is only firing for 0.7 s** and the exhaust is mostly out the back, so the risk is limited. However, the **inside of the motor mount** does get hot gases; that’s why we left a bit of motor sticking out and possibly will line the inside with a paper wrap. The **ejection charge** is a short burst of heat/flame; to protect the parachute, we have wadding (so that’s fine), and for the plastic, it might get a soot coating but should be okay (we must ensure that hole in the motor block so pressure can release, otherwise if sealed, the pressure and heat could melt things or burst the rocket). After firing, the motor itself will be hot (~maybe 50–70°C on the casing). It touching PLA could soften it. So ensure a **slight gap or insulation**: e.g., the motor mount could be a sleeve that is actually a bit loose, and we wrap the motor with masking tape to snug fit (the tape then acts as insulation too). Or print the motor mount with a thicker wall so the heat doesn’t transmit out quickly. Also, avoid having critical load-bearing parts right next to the motor if they could soften. Fortunately, by the time the rocket lands (~10 seconds after launch), things will have cooled somewhat. But if a second launch is attempted soon after, heat could build. So design with some **thermal tolerance**: e.g., motor retainer not relying on a thin PLA ring that could creep from heat – we prefer a tape solution or something not heat-sensitive.

Another safety design point: **nozzle blast and launch pad** – PLA at the tail will get the rocket exhaust flame on it for 0.7 s. Usually that’s fine but can char. We might consider applying a bit of aluminum tape or heat-resistant paint at the very rim if worried, but likely unnecessary for A8 motors. The key is that the motor’s thrust isn’t blocked. Our design ensures the motor sticks out, so the flame mostly doesn’t impinge on plastic. The **launch rod guide (lug)** should be high enough above the motor that exhaust doesn’t directly hit it (shouldn’t, since lug near CG ~ mid-rocket).

Finally, **post-print finishing**: we may do a light sanding on sharp edges (for safety and aero), and we’ll check for any printing defects (strings, blobs inside) that could interfere with chute. We’ll remove those. The structural design combined with quality printing and finishing will yield a rocket that is both robust and light.

- **Tolerancing for Assembly:** As we plan to iterate, we’ll ensure the design can be slightly adjusted without a complete overhaul. For instance, if a coupler is too tight, we can sand or trim – hence making couplers slightly longer than needed so trimming is possible. If the nose doesn’t fit, we can wrap tape for snugness. We intentionally incorporate the **0.010” clearance rule** everywhere because it’s easier to add a bit of tape to a loose fit than to deal with a too-tight fit that can crack things. We

also will consider how parts will be assembled: e.g., if gluing sections, provide **alignment marks or keys** perhaps (even a simple marker like a small notch to align fin can orientation to body tube so that the lug lines up between fins). We can add a tiny protrusion and slot as a key for proper alignment. Those need tolerance too but can be small (and if not fitting, we can cut them off and manually align).

In assembly, we will likely **glue the fin can to the body tube permanently** (for strength, unless the fit is super tight). We'll keep the nose removable. If something like the launch lug is separate, we'll glue that on carefully, using the printed fins as a guide to ensure parallel alignment. Because of tolerances, we may need to lightly sand coupler surfaces for glue to adhere well (glue holds better on a slightly roughened surface). Also, PLA doesn't bond strongly with some glues (CA works OK, epoxy works well but heavy). We might use a small amount of **epoxy or CA glue** on structural joints and scuff the surfaces first. This is outside of printing but influenced by how we design joint geometry (e.g., maybe include a slight recess or "glue groove" in couplers to hold adhesive).

By addressing all these structural and print considerations, we ensure the rocket can be manufactured reliably with the tools at hand, and that each iteration can be produced in ~10 hours or less (as noted, a full rocket print ~10 hrs ⁵⁵). This allows multiple trial prints within a 2-week period, which leads to our next section: rapid prototyping strategy.

5. Rapid Prototyping and Iteration Plan (2-Week Timeline)

- **Leverage Simulation Early:** Begin the design process in **OpenRocket (or similar)** to validate the basic geometry and flight performance before printing. Input the **Estes A8-3 motor data** (thrust curve, 2.5 N·s impulse, 3 s delay) to simulate the rocket's trajectory ¹⁵ ⁵⁶. Through simulation, iterate on fin size/shape, rocket length, and weight distribution quickly:
- Adjust fin dimensions until the simulated **stability margin is ~1.3–1.5** (with motor loaded) ².
- Check that the **launch rail exit speed ≥ 15.3 m/s** ³; if not, reduce mass or drag in the model.
- Observe the simulated **apogee time vs. delay**: ensure parachute ejection (at 3 s after burnout) is just after the peak altitude in the sim ². If the sim shows apogee at 2 s or 5 s, tweak the design (mass, drag) as discussed in §2.
- Confirm that **apogee altitude** is as high as possible given constraints. For instance, one can run an optimization: if making the rocket slightly longer (for stability) allows smaller fins (less drag), does altitude improve? Use the sim to guide these tradeoffs quantitatively. Look at altitude and also **landing speed** – ensure it's under ~5 m/s with the chosen chute size (OpenRocket can estimate descent rate; adjust parachute diameter input if needed to hit ~<5 m/s) ²⁸.

By simulating different configurations (e.g., compare 3 vs 4 fins, or different nose cone lengths/profiles), the team can converge on a promising design *before* printing. This saves time and material. Make note of the stability and performance metrics from OpenRocket for the final chosen geometry – these will serve as targets to verify in real tests.

- **Print Small Test Pieces First:** Once the design CAD is ready, perform some **test prints of critical sections** to validate dimensions and fit:
- **Motor Fit Test:** Print a short cylinder section of the motor mount (for instance, a 20 mm long piece of the fin can where the motor goes) and test insert an actual A8-3 motor. This will quickly reveal if

the diameter tolerance is correct (the motor should slide in with slight friction). If it's too tight/loose, adjust the CAD (e.g., change diameter or tolerance value) before printing the whole fin can.

- **Coupler Fit Test:** Similarly, print a small sample of a coupler and matching body tube (even just 10 mm high rings) to ensure the 0.25 mm clearance yields a snug fit that can slide. Test that you can insert and remove it by hand without cracking and that it aligns well. This prevents a scenario of printing full parts that won't go together.
- **Fin Profile Test:** If uncertain about fin printing orientation or strength, print one fin (or a portion of the fin and body) as a test. You could, for example, model a single fin attached to a bit of body tube and print it to see surface quality and strength. Then perform a simple bend test – does it feel sturdy enough? This can inform if you need to thicken the fin or add more fillet.
- **Shock Cord Attachment Test:** If the design has an internal hook or bar, try printing that section to see if the feature prints nicely (no excessive support needed, and it's strong). You might tie a piece of cord and tug to ensure it holds.

These tests can be done with low infill and coarse settings to be quick (since we mostly care about dimensional fit and basic strength). Each test might take 30–60 minutes instead of a whole rocket taking 8+ hours. By investing a half day in such tests, the team can avoid wasting a full day on a flawed full print.

- **Rapid Iteration of Prints:** Plan for multiple iterations within the 2-week schedule – **start printing early**. According to project info, a full rocket can be printed in ~10 hours ⁵⁵, so if printing overnight or on multiple printers, you could make several versions. The mindset should be: **print, test, learn, improve, re-print**.
- **Version 1 (Week 1):** Print an initial prototype of the complete rocket (after doing the small tests above). Use default settings (20% infill, recommended wall thickness) and perhaps slightly heavier settings if worried (better a bit heavy but robust for first test). Once printed, **inspect and assemble** it. Does everything fit? (Nose to body, etc.) Does the motor insert fully and stop at the block correctly? Pack the parachute and do a **shake test** – hold the rocket upside down and shake lightly, the nose shouldn't drop out on its own. Then do an **ejection ground test** if possible: Remove the motor's black powder ejection charge from an A8-3 (or use an ejection test tool) – you can simulate by putting a little wadding and using a CO₂ cartridge or just blowing air to see if the chute comes out easily. If you have a spare motor, you could even mount the rocket (without engine igniting) and just fire the ejection to see deployment. This will reveal any issues like nose too tight, chute hanging up, etc., which you can address in CAD for V2.
- **Version 2 (Mid Week 2):** Incorporate fixes: e.g., sand or adjust dimensions that were off, lighten any areas if the first one was too heavy, etc. Possibly, if the first flight (if you did one) showed needed tweaks (maybe fin too small or rocket marginally stable), adjust accordingly (increase fin area a bit or add small nose weight as last resort – if adding nose weight, you could design a small cavity in nose to place clay). Print the second rocket. This one should be closer to final. If the first one was good enough to test fly, do so; if not, definitely plan a **test launch for version 2**. Preferably, schedule a safe test launch before the final launch day to see actual performance. Measure or observe the apogee (even rough estimate). Did it eject near apogee? If it looked like ejection was late or early, that gives insight: e.g., if chute popped well after peak, maybe rocket could be lighter or have more drag. If rocket weathercocked a lot (turning into wind), it might be too stable or leaving rail slow – consider slight adjustments (reduce fin area, or ensure full 1 m rail and low wind).
- **Final Version (End of Week 2):** With lessons learned, print the refined final rocket for the launch. Ideally, this version has all known issues fixed: great fits, reliable recovery, and dialed-in stability. This print can be done with a bit more care for surface quality (maybe slightly finer layer or more infill if

needed for strength) now that we're confident in geometry. If painting or finishing is desired, we'd do that after this print.

Each iteration, document the changes – this helps ensure you don't accidentally revert a fix. For example, if V1 had nose too tight, note to sand or adjust in CAD; if V2 had perfect nose fit but fins still a bit weak, maybe fillet more in V3. Keeping a log will make the process systematic rather than trial-and-error. The project encourages iterative design ⁵⁵, so using the full two weeks for design-print-test cycles will yield a much better rocket.

- **Prototyping Practical Tips:** During the iteration process, use the available resources:
- The **3D printer settings** can be tweaked to save time. For instance, for prototype prints, use a larger layer height (0.25 mm) to cut print time, or lower infill if structure isn't being tested. For the final print, revert to finer settings for strength and finish.
- If something is consistently problematic to print (say the fins with support), consult an instructor or experienced peer – perhaps they have a slicing trick like “support blocker” or a different orientation. Don't waste too many tries on a print failure; instead, modify the design or settings.
- Use the allowed materials for quick fixes in prototypes: e.g., masking tape to temporarily hold parts or to adjust fit (wrap tape on a loose coupler to see how much needs changing in CAD). Hot glue can tack pieces together to simulate final assembly, then be popped off for adjustment.
- **Mass tracking:** Weigh the rocket components as you go. Compare against simulation mass. If heavier, decide if it's acceptable or if you should modify the design (maybe you printed walls thicker than needed?). We can also intentionally try a lightweight print (like lower infill, thinner wall) and a heavier robust print, and flight-test both to see altitude differences. In a competition for max altitude, those grams matter.
- **Center of gravity testing:** Once a prototype is assembled with motor (unlit) and parachute in, balance it on a ruler edge to find CG. Mark it and compare to CP from OpenRocket. Ensure margin is as predicted. If CG is too far back (stability low), consider adding a tiny bit of clay to nose for test launch to be safe, then later fix by design (e.g., slightly longer nose or fin tweak).
- **Swing test:** As a quick qualitative stability check, do a **swing test**: attach a string at the rocket's CG and swing it in a circle overhead. The rocket should naturally point nose-forward along the direction of travel (like a weather vane). If it flies backwards or tumbles, it's unstable. This test isn't perfectly reliable, but if it passes, it's a good sanity check before actual launch.
- **Launch day readiness:** Ensure you have spares or repair tools. For example, if fins tend to crack, print a spare fin can or have epoxy to fix. If the nose fit is iffy, bring tape or sandpaper to adjust at the field. Since our rocket is 3D-printed (brittle), handling carefully is important; but with parachute landing it should survive fine. Rapid prototyping means we might have multiple rockets – it's actually good to have a backup rocket assembled in case one breaks on launch day.
- **Final Verification:** The result of iterative design will be a rocket that satisfies:
 - **Stability 1.1–2.5 calibers** (we'll aim ~1.5) in final sim and as-built, checked by measurement ².
 - **Ejection after apogee** in sim and (hopefully) observed in test – meaning we succeeded in matching the delay to flight profile ².
 - **Off-rail velocity ≥ 15.24 m/s** confirmed by sim and by the fact the test flight was straight (if it left slower, it would likely show a pronounced arc or instability) ³.
 - **Mass ≤ 3 oz (85 g)** – we will weigh the final rocket with motor to ensure this ⁵⁷. If it's, say, 80 g, we're good; if it came out 90 g, we'd need to quickly drill some holes or strip weight last-minute.

- **Parachute fits and deploys** – ground test the packed chute one last time: plug in a motor (or dummy motor) with wadding and ensure we can pop the nose off with a gentle push, parachute comes out smoothly. Also double-check **shock cord knots and attachments** are secure (give them a firm tug test).
- **Launch prep checklist:** put some baby powder on the chute (helps deployment), insert proper amount of wadding, install motor with a bit of tape for snugness, align on launch rail (make sure lug slides freely). The design's proper execution will make all these steps easy and reliable.

Throughout the 2-week development, the team will have learned from each prototype – gaining confidence in CAD adjustments, print settings, and understanding how **mass, stability, and drag trade off** in real life. By final launch day, we'll have a refined rocket design that meets the class's specs and is tuned for maximum altitude given the A8-3's limited power ²⁴. This process embodies **rapid prototyping and iterative design**: building, testing, and improving quickly to achieve an optimal result.

Sources:

- Estes A8-3 Motor Specifications – Estes Rockets (Total impulse 2.50 N·s, burn time ~0.7 s, max thrust ~10 N, 3 s delay, recommended max liftoff weight 85 g) ¹⁵ ¹⁷
- Thrust Curve and Performance Data – ThrustCurve.org (A8 motor average thrust ~3.2 N, 0.7 s burn) ²⁵ ⁵⁸
- Project Assignment Guidelines – *E1RK 3D Printed Rocket, Spring 2025* (design requirements: stability 1.10–2.50 calibers, >50 ft/s off rail, <3 oz mass, ejection after apogee 3 s delay, etc.) ² ³. Includes printing tips ($\leq 8''$ parts, $\geq 1/16''$ walls, tolerances ~0.010") ¹² ⁴ and construction checkpoints (shock cord anchors ≥ 3 mm, motor block with vent hole, parachute volume ≥ 15 cm³, lug dimensions) ³⁸ ⁴¹.
- Aerodynamic Design References: MIT Rocket Team Wiki (nose cone shape for subsonic minimum drag: use curved/ogive shapes) ⁶, Apogee Rockets research (elliptical nose cone yields lowest drag in subsonic tests) ⁹. Fin stability vs. drag tradeoff (more fins = more stability but more drag), and importance of airfoiled fins (reduces drag, improves stability) ¹¹. These principles guided the body and fin design for optimal altitude.

¹ ¹⁵ ¹⁷ ¹⁹ ²⁶ ⁵⁶ **A8-3 Engines - Standard Model Rocket Engine 3-Pack | Estes Rockets**
<https://estesrockets.com/products/a8-3-engines?srltid=AfmBOoo4drpuh4tZ9LzuQdv1ITXsKZo7koXkbzii7XAntzQXZgiYwRzg>

² ³ ⁴ ⁵ ⁷ ⁸ ¹⁰ ¹² ¹³ ¹⁴ ¹⁶ ¹⁸ ²⁴ ²⁷ ²⁸ ²⁹ ³⁰ ³¹ ³² ³³ ³⁴ ³⁵ ³⁶ ³⁷ ³⁸ ⁴⁰ ⁴¹ ⁴² ⁴³ ⁴⁴
⁴⁵ ⁴⁶ ⁴⁷ ⁴⁸ ⁴⁹ ⁵⁰ ⁵¹ ⁵² ⁵³ ⁵⁴ ⁵⁵ ⁵⁷ **2.4 3DPR Assignment.pdf**
file:///file_00000000794461fa802e39a686343c0e

⁶ **How to: Choose a Nose Cone - MIT Rocket Team - MIT Wiki Service**
<https://wikis.mit.edu/confluence/display/RocketTeam/How+to+%3A+Choose+a+Nose+Cone>

⁹ **apogeerockets.com**
https://www.apogeerockets.com/downloads/Drag_of_Nose_Cones.pdf

¹¹ ³⁹ **Bubble, 3D Printed Model Rocket : 9 Steps (with Pictures) - Instructables**
<https://www.instructables.com/Bubble-3D-Printed-Model-Rocket/>

20 21 22 23 **Peak of Flight Newsletter : Apogee Rockets, Model Rocketry Excitement Starts Here**
<http://www.apogeerockets.com/Peak-of-Flight/Newsletter594>

25 58 **Estes A8 • ThrustCurve**
<https://www.thrustcurve.org/motors/Estes/A8/>